Matt Schwennesen

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Education

Aug 2023 – · · · · **University of Wisconsin – Madison,** Pursuing Ph.D. in Computer Sciences.

Aug 2023 – Dec 2025 **University of Wisconsin – Madison,** M.S in Computer Sciences

Aug 2019 – Dec 2022 **Michigan Technological University,** B.Sc. Computer Science
Minor in Mathematical Sciences, GPA: 4.00

Teaching Experience

Jun 2026 – Aug 2026 **Graduate Student Instructor,** University of Wisconsin – Madison.
CS 400 Programming III

Jan 2026 – May 2026 **Teaching Assistant,** University of Wisconsin – Madison.
CS 400 Programming III

Jan 2024 – Dec 2024 **Head Teaching Assistant,** University of Wisconsin – Madison.
CS 400 Programming III

Aug 2023 – Dec 2023 **Teaching Assistant,** University of Wisconsin – Madison.
CS 400 Programming III

Aug 2022 – May 2023 **Teaching Assistant,** Michigan Technological University.
CS 3411 Systems Programming

Jan 2023 – May 2023 **Lab Instructor,** Michigan Technological University.
CS 1121 Introduction to Programming I

Jan 2021 – Dec 2022 **Learning Center Coach,** Michigan Technological University.

Conferences

Nov 22 2024 **Midwest Programming Languages Summit**, University of Chicago

Jun 17–21 2024 **NetSci 2024**, Quebéc City, Canada.
Presented during *Software Tools for Network Science* tutorial on cross package network analysis.

Jun 3–13 2024 **Oregon Programming Languages Summer School**, Boston University, Massachusetts.
Types, Semantics and Applications

Research

Aug 2026 – · · · · **Asynchronous Choreographic Programming with Network Failures**
Work advised by Ethan Cecchetti.

- Exploring the relaxation of Choreographic Programming model assumptions to handle asynchronous communications in the face of networks reordering and dropping messages.

Research (continued)

Feb 2025 – · · · · **Pollux: Verified Updates to Data Descriptor Formats**
Work advised by Tej Chajed.

- Currently researching verifiably compatible updates to Protocol Buffer descriptors and JSON messages.
- Exploring formal definitions for update compatibility and constructing a verified compatibility checker.

Aug 2024 – Jan 2025 **Grackle: Proof-Instrumented Marshaling & Unmarshaling.**
Work advised by Tej Chajed.

- Investigated techniques for automating repetitive Rocq proofs using Goose and Perennial.
- Implemented a go program generating marshaling and unmarshaling code for protobuf messages and a Rocq proof of correctness.

May 2022 – Aug 2022 **Locality Sensitive Hashing of Polygons.**
Research Experiences for Undergraduates – Marquette University.

- Researched uses of locality sensitive hashing to approximate nearest neighbor searches over polygons.

May 2021 – Aug 2021 **Asymmetric Traveling Salesperson Approximation.**
Google Summer of Code – NetworkX.

- Worked with NetworkX to implement approximate asymmetric traveling salesperson algorithm from original publication.

Research Publications

Conference Proceedings

1. A. Subedi, **M. Schwennesen**, O. Durowoju, and S. Puri, "PolySS: A Compact Filter-and-Refine System for Efficient Approximate Polygon Similarity Search", to appear in *2026 IEEE International Conference on Big Data*, Dec. 2026.
2. B. A. M. K., S. Puri, A. Soman, **M. Schwennesen**, and S. K. Prasad, "ShapeToVec: Encoding Polygonal Shapes with Extreme Area Variability for Effective Approximate Jaccard Similarity Queries", in *2025 IEEE International Conference on Big Data (BigData)*, Dec. 2025, pp. 530–539. DOI: 10.1109/BigData66926.2025.11401899 Accessed: Apr. 1, 2026.

Service

Jan 2026 - · · · · **President, Student Chapter of the Association for Computing Machinery**
Jan 2025 - · · · · **Contract Enforcement Committee, Teaching Assistants Association**
Aug 2026 **Artifact Evaluation Committee, SPLASH 2026**

Skills

Mathematics	Proof tactics, mechanized proofs, separation logic, programming languages, graph theory, combinatorics, linear optimization, algorithm design, complexity theory
Programming	Rocq, Lean, Go, Python, C, C++, Java, Nix, Haskell, F*, Lisp, R
Misc.	Academic research, teaching, LaTeX typesetting.